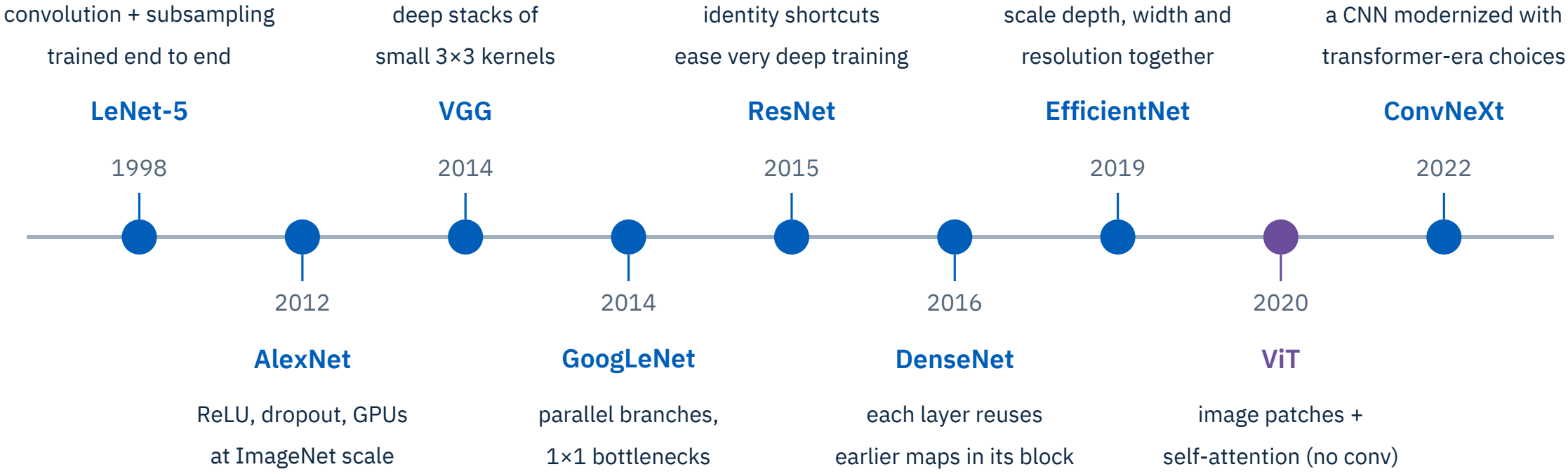
convolution + subsampling
trained end to end
LeNet-5
1998
2012
AlexNet
ReLU, dropout, GPUs
at ImageNet scale
deep stacks of
small 3×3 kernels
VGG
2014
2014
GoogLeNet
parallel branches,
1×1 bottlenecks
identity shortcuts
ease very deep training
ResNet
2015
2016
DenseNet
each layer reuses
earlier maps in its block
scale depth, width and
resolution together
EfficientNet
2019
2020
ViT
image patches +
self-attention (no conv)
a CNN modernized with
transformer-era choices
ConvNeXt
2022